

Wesley Deklich

408-647-0040 • wesleydeklich1@gmail.com •  wesleydeklich •  wesleydeklich

EDUCATION

University of Illinois Urbana-Champaign
B.S. in Computer Science & Economics, Statistics Minor
Relevant Coursework: Data Structures & Algorithms · Systems Programming · Database Systems · Software Engineering

Expected Graduation: May 2027
GPA: 3.97/4.0

EXPERIENCE

Aisera *Software Engineering Intern*
June 2025 – August 2025

- Built scalable RAG pipeline with ontology-driven search, improving query accuracy by **18%** through strategic integration of entity hierarchies with production backend services.
- Designed clustering microservices for enterprise knowledge organization, enhancing retrieval precision by **22%** and reducing system noise through distributed architecture.
- Optimized vector database performance for multi-million record datasets, reducing query latency from **120ms to 45ms** while maintaining 99.9% system uptime.

RingCentral Inc *Software Engineering Intern*
November 2022 – January 2023

- Optimized cloud-native video infrastructure serving **1M+ users** by implementing compression algorithms and scaling distributed systems, improving video quality and reliability.
- Built real-time analytics platform with Python microservices and automated pipelines, processing video metrics to identify bottlenecks and optimize throughput.
- Developed interactive web applications with React and D3.js featuring real-time visualization, increasing user engagement by **20%** through responsive design.

NASA *Software Engineering Intern*
May 2022 – October 2022

- Engineered data processing systems handling **10,000+ meteorite records** using Python, implementing efficient ETL pipelines and database optimization for scientific applications.
- Designed and deployed classification system with TensorFlow, achieving **15%** precision improvement through feature engineering and automated validation workflows.
- Built scalable analysis platform processing **5,000+ samples** with **20%** accuracy improvement, implementing comprehensive error handling for production deployment.

PROJECTS

Investor Analysis Dashboard *FastAPI · React · MongoDB · Docker · AWS*
October 2024 – Present

- Built full-stack application with FastAPI backend and React frontend, implementing real-time sentiment analysis with comprehensive error handling.
- Designed MongoDB architecture supporting **500,000+ records** with **99.9% uptime**, implementing efficient indexing and query optimization.
- Deployed containerized application on AWS with CI/CD pipeline, implementing load balancing and auto-scaling for production reliability.

TECHNICAL SKILLS

Languages: Python · C/C++ · Java · Go · JavaScript/TypeScript · SQL · Bash
Backend & Web: FastAPI · Flask · Node.js · React · GraphQL · REST APIs
Databases: PostgreSQL · MongoDB · Redis · Elasticsearch · Vector DBs
Cloud & DevOps: AWS · Docker · Kubernetes · CI/CD · Terraform · Linux
ML & Data: PyTorch · TensorFlow · NumPy · Pandas · Kafka · RabbitMQ